

Chapter 7: SVM-based Speaker Classification

This Chapter was based on a class project completed by Ansen Herrick. All materials, including project codes and results, can be found at the book GitHub repo: https://github.com/AVHBAC/SVM_Speaker_Classification.

This project was chosen because it is believed that speaker classification is a useful field that has many applications. Speaker recognition could be used in a security system to ensure that only certain subjects can access certain areas, whether that be in real life with a deployed system on a door lock system, or for use in online data storage systems, where some directories could be locked behind a speaker verification system. It could also be used in conjunction with other biometric systems for added security, for example, a system that uses a fingerprint and speaker verification network. The dataset selected for this project (VoxForge) is very useful for this task because it contains many audio samples, which all contain recordings of the same prompts, of many unique users. This dataset will provide a lot of diverse accents and speaking patterns that the Support Vector Machine (SVM) can use to classify. Various input feature vectors will be used, including MFCCs, which capture the spectral properties in a speaker's voice adjusted to the range of human hearing. SVM models require an enrollment process for specific users, where voice samples from a single subject will be collected and labeled. SVM will learn the unique characteristics of this certain speaker. Then, when a new voice is presented, SVM will determine how similar the voice features are to the desired class, and based on the similarity, the system will be able to tell how confident it is that the new voice is the same as the desired voice. If this approach is successful, the community will be benefited from a secure system for speaker verification. This can be used in many applications as explained above. The System could then eventually be modified for more complex tasks, such as using speaker verification for data encryption and storage.

7.1 Motivation

SVM is a supervised learning algorithm used primarily for classification and regression. They are very effective at separating multidimensional data. Usually, SVM models are used to classify two different classes, but in this case, an SVM Model with a 'One vs. Rest' technique will be used to distinguish one class from the rest. This is achieved by finding a maximum separating hyperplane between classes using the provided features. The model can analyze data in the feature space in order to find an effective separating hyperplane. For the case of speaker classification, the SVM model should be very effective at finding an effective separation plane in order to distinguish classes from one another.

7.2 Dataset

The dataset used in this study was gathered from the VoxForge Dataset, a famous open-source speech dataset that dates back to 2006, where users are able to upload their own speech files for free use in deep learning algorithms and machine learning. The dataset for this model consists of speech data from 10 different users. These users were selected using a singular criterion, the amount of data each user had uploaded. Each user or 'class' in the dataset has at least 250 voice samples in order to effectively train the SVM model. The dataset consists of 4779 individual voice samples, split between training and testing. Each of the collected samples ranged from 3-15 seconds long and were recorded at a sample rate of 48kHz.

7.2.1 Data pre-processing

Various pre-processing steps were applied to the data in order to ensure the most effective model training. As said above, there are 10 different speakers or 'classes' used in the dataset. Table 7.1 outlines the number of voice files collected from each user and how the training and testing dataset was split.

The data was then de-noised using a noise reduction in python called 'noisereduce.' The threshold set for this operation was 0.8, this value was used because it significantly reduces the noise in the audio files while still preserving a majority of the original sound quality. Most samples recorded where from the years 2007-2009, so noise was very prominent.

7.3 Model Training

SVM model was trained using an RBF (Radial Basis Function) kernel using One vs. Rest classification. A kernel is a function that is capable of computing similarity between two data points in feature space. This can allow the SVM model

User	Training Files	Testing Files	Total Files	Training/Testing Split
akiplaner	468	52	520	90/10
bonzer	351	39	390	90/10
camdixon	537	60	597	89.9/10.1
corno	340	38	378	89.9/10.1
doublesfrogs	262	28	290	90.3/9.7
farmerjack	548	62	610	89.8/10.2
k	333	37	370	90/10
pcsnpy	612	68	680	90/10
ralfherzog	296	38	334	88.6/11.4
rortiz	549	61	610	90/10
Total	4296	483	4779	89.9/10.1

Table 7.1: Number of Files per User in the Dataset

to handle non-linear relationships. The RBF kernel was chosen because it works well in a variety of problems and is effective even with no prior knowledge about the data structure. [1] The RBF kernel assigns a higher similarity score to points that are close in the feature space, while applying a lower similarity score to points that are far apart. The SVM model was fit to the Training Dataset; the model took 9 minutes to train on average across multiple sessions.

7.4 Results

7.4.1 Accuracy

Following the training of the model, the SVM was used to predict the speakers class with the testing dataset. Across 483 testing audio files, the model reached an accuracy of 97 percent with 3 MFCC Coefficients. Some users or ‘classes’ had a testing accuracy of 100 percent, while the lowest accuracy for any user was 82 percent. This data can be effectively visualized using a Confusion matrix, which is displayed in Figure 2.1.

7.5 Statistical Tests

7.5.1 ANOVA

An ANOVA feature test was applied to the feature set after training to determine which of the extracted features were most significant in classifying the subject. ANOVA stands for Analysis of Variance. It works by comparing the variance between groups based on the target variable. The ANOVA returns an F-value, where a high F-value indicates that the specific feature plays a significant role in distinguishing between classes. The F-values for all of the features are displayed in Figure 7.2, and the descriptions of each feature are shown in Table 7.2, where each MFCC Coefficient and the associated operation are described. By observing Figure 7.2, it can be seen that the top 5 features for correctly classifying speakers were index

Feature Index	Description	Feature Index	Description
0	Coefficient 1 Mean	11	Coefficient 2 Kurtosis
1	Coefficient 1 Median	12	Coefficient 2 Maximum
2	Coefficient 1 Std Dev	13	Coefficient 2 Minimum
3	Coefficient 1 Skewness	14	Coefficient 3 Mean
4	Coefficient 1 Kurtosis	15	Coefficient 3 Median
5	Coefficient 1 Maximum	16	Coefficient 3 Std Dev
6	Coefficient 1 Minimum	17	Coefficient 3 Skewness
7	Coefficient 2 Mean	18	Coefficient 3 Kurtosis
8	Coefficient 2 Median	19	Coefficient 3 Maximum
9	Coefficient 2 Std Dev	20	Coefficient 3 Minimum
10	Coefficient 2 Skewness		

Table 7.2: Description of MFCC Feature Labels

18, The kurtosis of MFCC Coefficient 3, index 7, The Mean of Coefficient 2, index 6, The Minimum of Coefficient 1, index 2, the Median of Coefficient 1, and index 8, the Median of Coefficient 2. In order to better visualize how the variance of these features is separated, Figures 7.3 through 7.7 display the variance distribution across speakers.

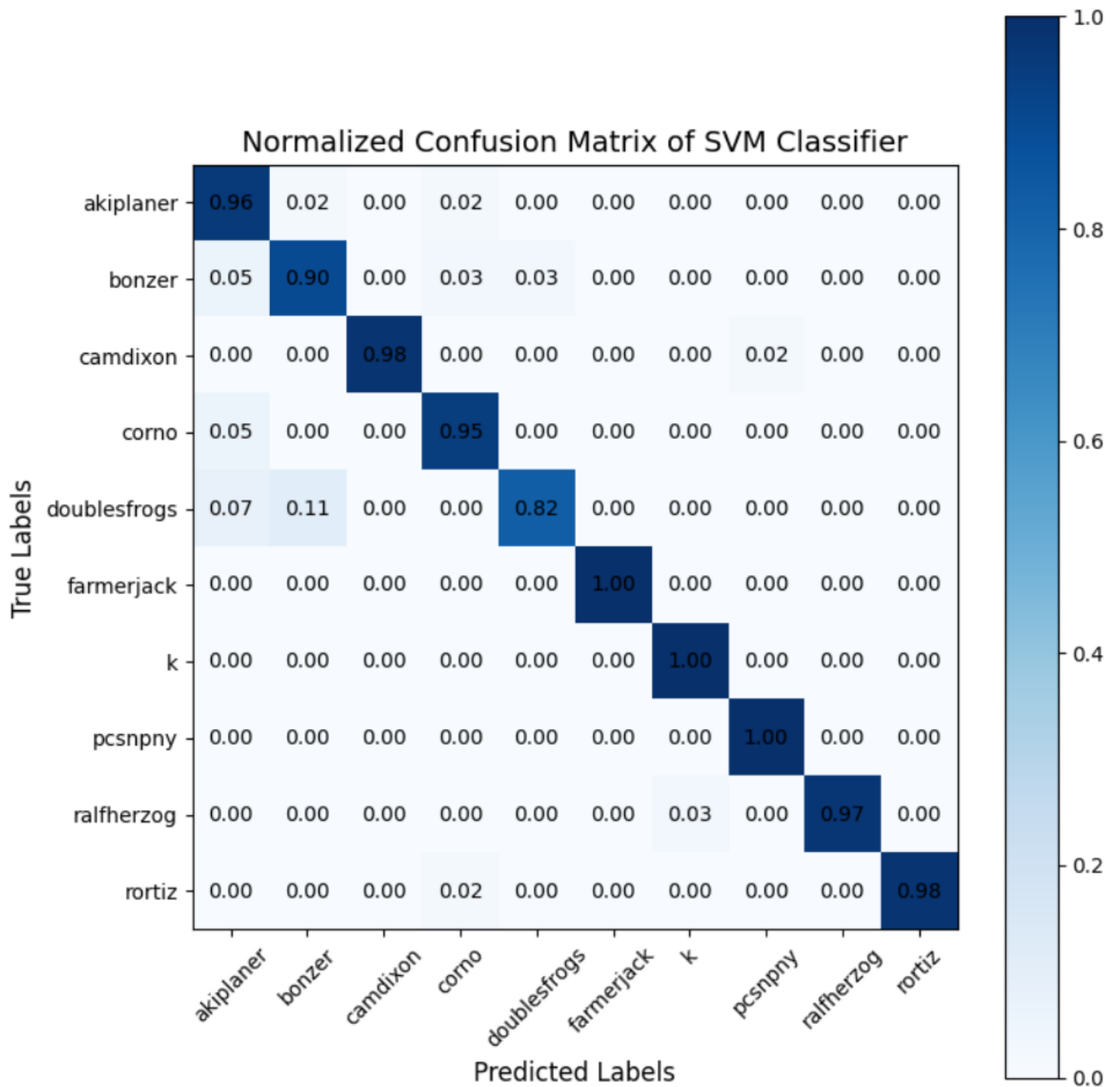


Figure 7.1: Normalized Confusion Matrix of SVM Classifier

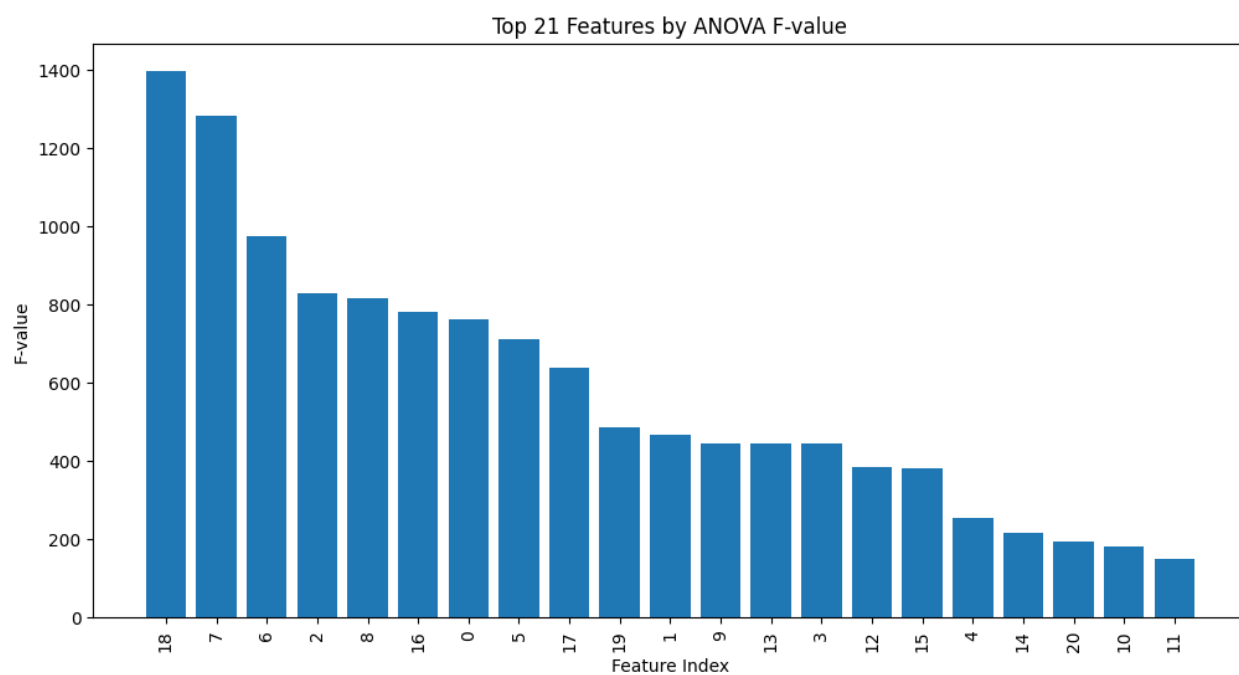


Figure 7.2: F-values for extracted features

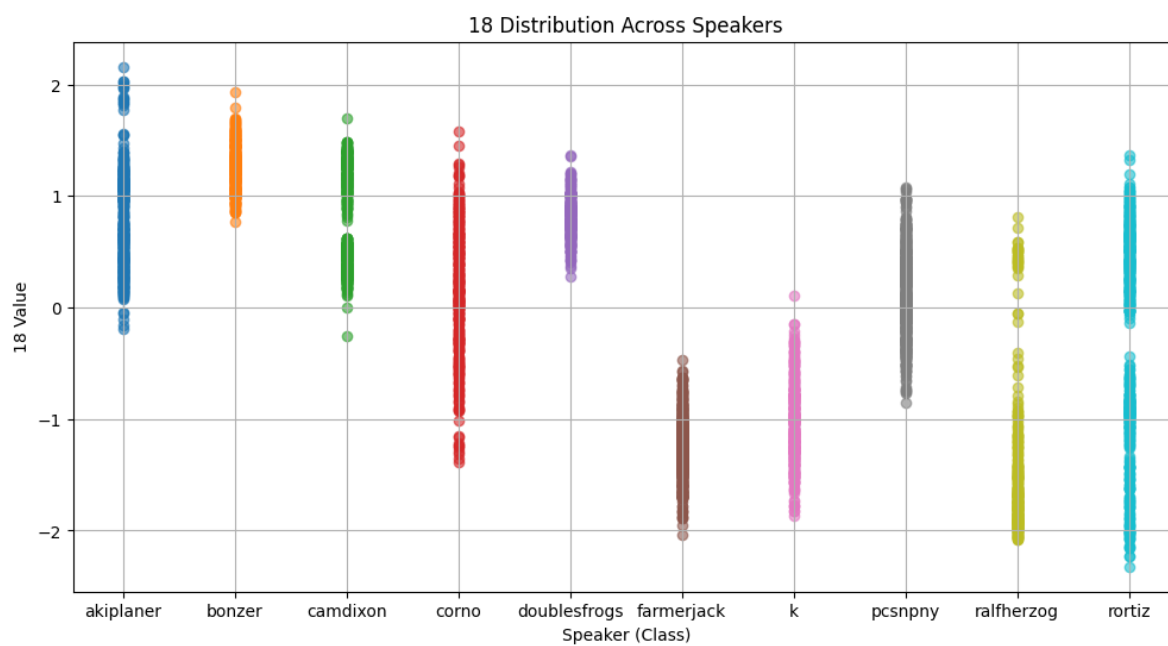


Figure 7.3: Distribution of the Kurtosis of MFCC Coefficient 3

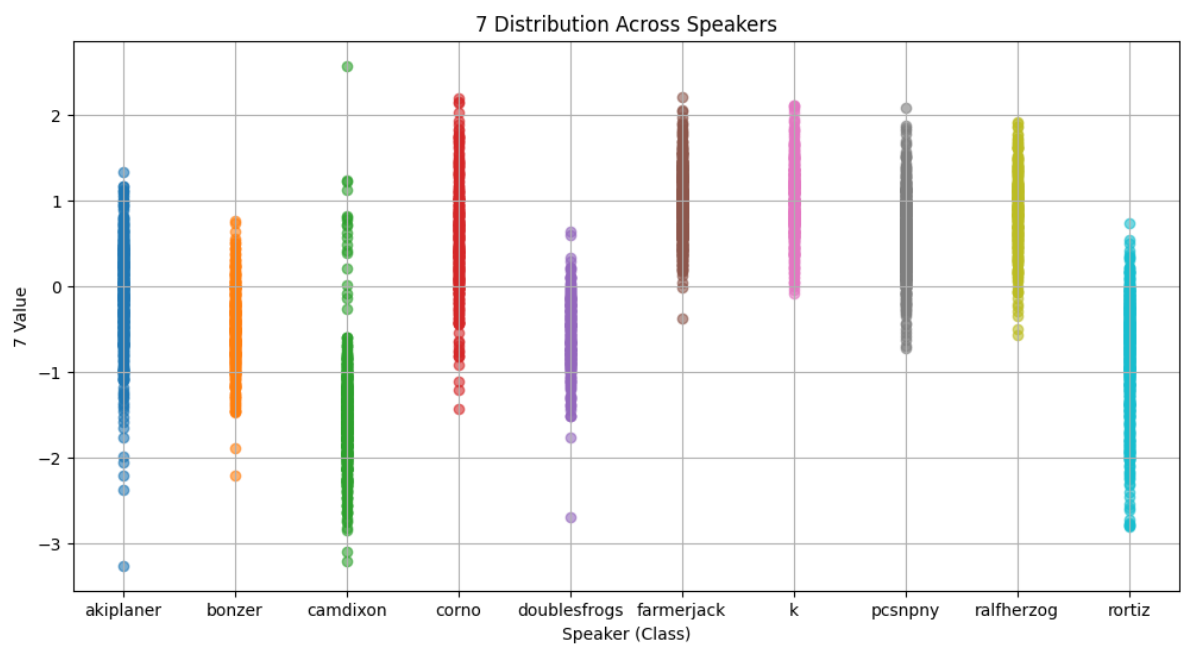


Figure 7.4: Distribution of the Mean of MFCC Coefficient 2

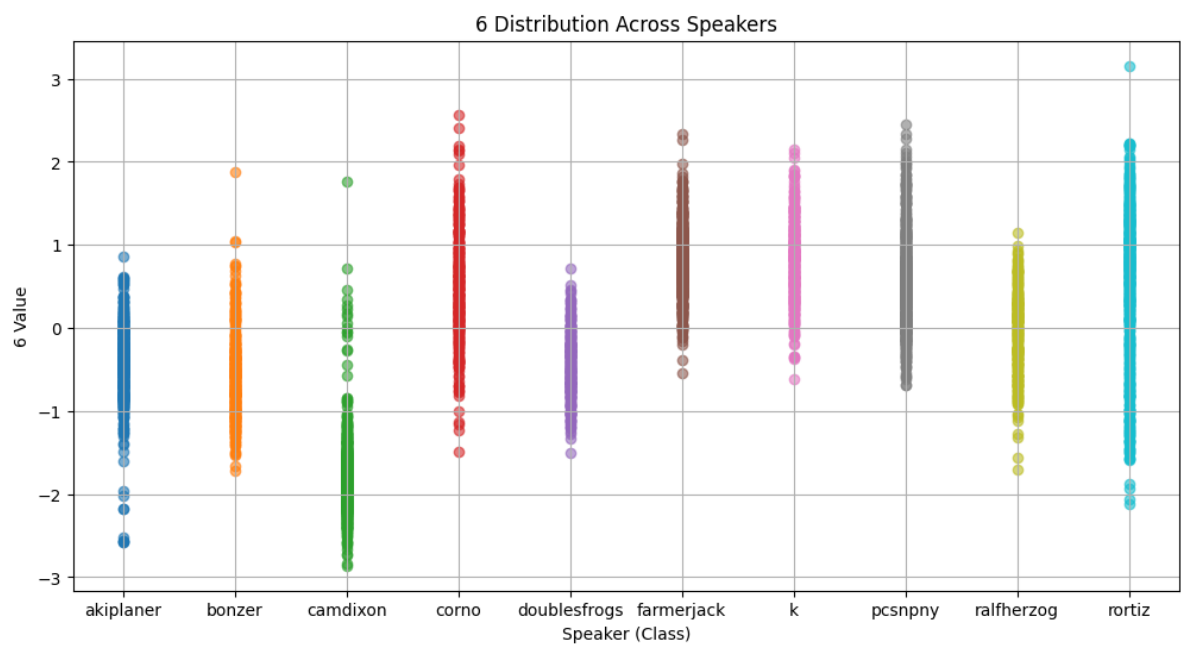


Figure 7.5: Distribution of the Minimum of MFCC Coefficient 1

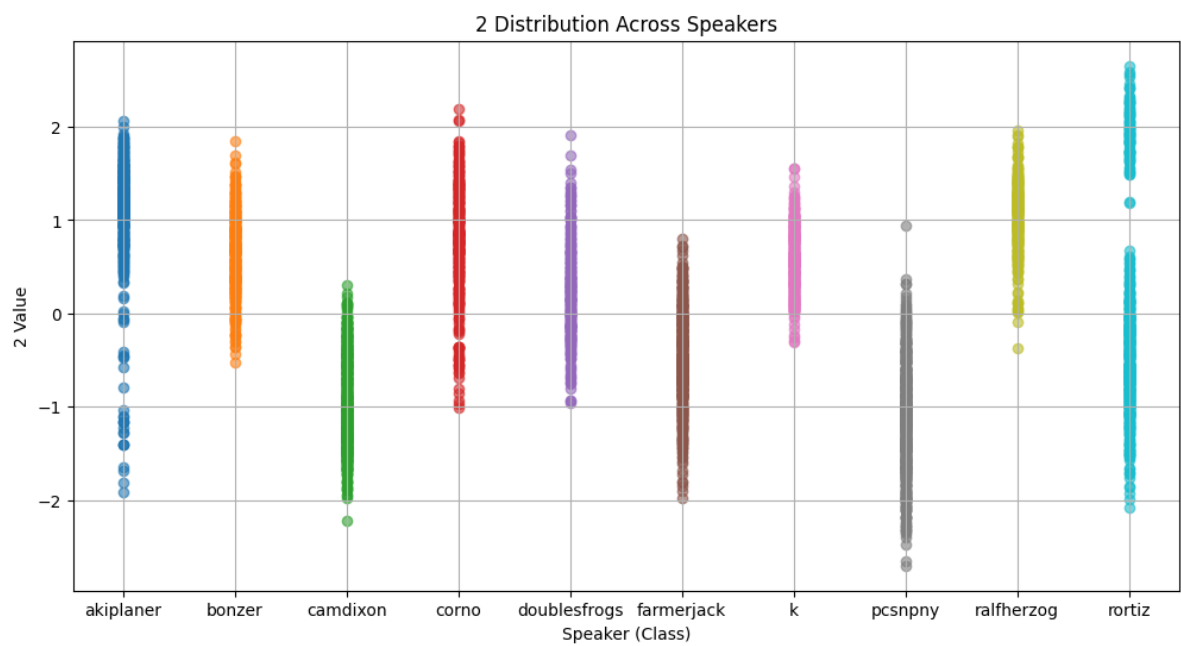


Figure 7.6: Distribution of the Median of MFCC Coefficient 1

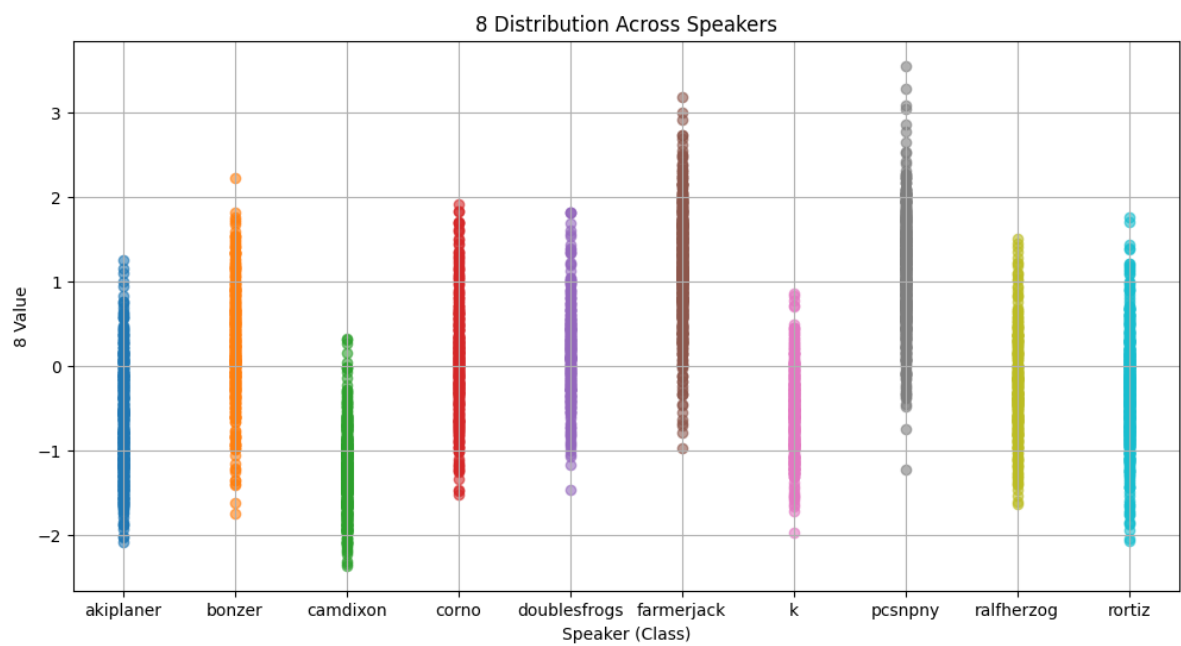


Figure 7.7: Distribution of the Median of MFCC Coefficient 2

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
0.96162791	0.9371362	0.84866123	0.94295693	0.88824214

Table 7.3: Accuracies for each Fold during Cross-Validation

Figures 7.3 through 7.7 display the variance of the designated feature extracted. The SVM is optimized to notice the difference in these distributions and can use these differences to accurately classify audio files.

7.5.2 K-Fold Cross Validation

A K-Fold Cross Validation test was used to evaluate the performance of the SVM model. A K-Fold test uses 'k' different "folds" to compare how the model is affected by different groups of data. Each fold comprises an equally split portion of the dataset. The model will then train itself on k-1 out of k datasets and use the remaining dataset for training. A K-Fold test can show how the model might perform when used with unseen data. A 5-fold cross validation test was used to evaluate the performance of the model. The accuracy of each fold is displayed in Table 7.3. The Mean accuracy of all the folds was 92 percent, with a standard deviation of 0.04. The average accuracy of the K-Fold test was observed to be lower than the final model accuracy on the test set. This could mean a variety of things, such as the model is over fitted to the test set, or the data splits in the cross-validation are more challenging than the original training set. A lower accuracy in cross-validation is also common in smaller datasets because the variance can be higher due to limited samples in each fold.

7.5.3 Chi-Squared Test

A chi-squared test was used to determine whether the observed results of the model differ significantly from the expected frequencies, or the null hypothesis. In this case, the expected frequencies were set to a normal, random distribution. So the results of the Chi-Squared test will determine how much better the SVM model is compared to random classification. The Chi-Squared test provided a value of 3981.8 and a p-value of $8.771e-5$. In this case, the null hypothesis suggests the predictions are made randomly, so a high Chi Squared value and a low p-value show that the model isn't behaving randomly.

7.6 Related Studies

Spoken Speaker identification based on Gaussian Mixture Models is an article authored by Abhijeet Kumar outlining a speaker identification system using the VoxForge dataset. This study, instead of using SVM for classification is using GMM, which is a statistical clustering model, rather than a classification model. The study consists of 34 different speakers speaking five different speech utterances. The dataset differs from this dataset because the users are all saying the same thing, and there are many more classes. This study also extracts 40 MFCC features, 20 of which are directly from the MFCC and 20 are derivatives of the MFCC. After training the Gaussian Mixture Model, it was tested on 5 different speech utterances from the 34 different users. This model reached an accuracy of 100 percent. The performance of this model was likely better than the SVM model's performance for a couple of reasons. First, the fact that the model was trained on the same speech utterances means that the model could pick up distinct differences in the speakers' voices, and would ignore any differences found from the different users speaking different words. The GMM Model also utilized more MFCC features than this model, which could've made it more effective. [2]

7.7 Conclusion

Speaker Recognition has been a difficult task for computers for a long time, but this article discusses the creation of a Support Vector Machine Model to solve that problem. The Model was trained on over 4,000 speech samples that were denoised and segmented into 20-second intervals. Each speech sample had features extracted using MFCC Coefficients because they proved to be very effective in capturing individual speaker characteristics. After training, the model reached an accuracy of 97 percent. By applying an ANOVA statistical test to the features, the most significant features extracted for identifying each speaker can be observed. Two statistical tests were also applied to the models performance, K-Fold cross-validation and a Chi-Squared test, in order to verify the legitimacy of the model's accuracy. During cross validation the model received an accuracy of 92 percent, and even though this accuracy is lower it can still be explained. Finally, other studies were able to reach higher levels of accuracy using other techniques, but the SVM model trained in this paper is still very effective for speaker classification. All codes used to create the model in this chapter will be available on GitHub at SVM-Speaker-Classfier.

Bibliography

- [1] S. Eskandar. (2023) Introduction to rbf svm. Accessed: 2024-11-26. [Online]. Available: <https://medium.com/@eskanndar.sahel/introduction-to-rbf-svm-a-powerful-machine-learning-algorithm-for-non-linear-data-1d1cfb55a1a>
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